$$
\begin{aligned}
& C_{\phi D}^{(21)\,(12)} \to \frac{1}{16\pi^2} \Big(-\frac{1}{9} g_2^4 \mathrm{LF}_{3,0}[m_2] - \frac{1}{6} g_2^4 \mathrm{LF}_{4,-1}[m_2] + \frac{8}{45} g_2^4 \mathrm{LF}_{5,-2}[m_2] + \\
& \left(-\frac{1}{4} g_2^2 \overline{y_e}^{pr} y_e^{pr} + \frac{5}{72}\sum_p g_2^4\right) \mathrm{LF}_{3,0}[m_{\tilde l}^p] + \left(\frac{1}{4} g_2^2 \overline{y_e}^{pr} y_e^{pr} - \frac{1}{48}\sum_p g_2^4\right) \mathrm{LF}_{4,-1}[m_{\tilde l}^p] - \\
& \frac{2}{45}\sum_p g_2^4 \mathrm{LF}_{5,-2}[m_{\tilde l}^p] + \left(-\frac{3}{4} g_2^2 (\overline{y_d}^{pr} y_d^{pr} + \overline{y_u}^{pr} y_u^{pr}) + \frac{5}{24}\sum_p g_2^4\right) \mathrm{LF}_{3,0}[m_{\tilde q}^p] + \\
& \left(\frac{3}{4} g_2^2 (\overline{y_d}^{pr} y_d^{pr} + \overline{y_u}^{pr} y_u^{pr}) - \frac{1}{16}\sum_p g_2^4\right) \mathrm{LF}_{4,-1}[m_{\tilde q}^p] - \frac{2}{15}\sum_p g_2^4 \mathrm{LF}_{5,-2}[m_{\tilde q}^p] - \\
& \frac{1}{18} g_2^4 \mathrm{LF}_{3,0}[\tilde\mu] - \frac{1}{12} g_2^4 \mathrm{LF}_{4,-1}[\tilde\mu] + \frac{4}{45} g_2^4 \mathrm{LF}_{5,-2}[\tilde\mu] + \frac{2}{3} g_2^4 \mathrm{LF}_{2,1,0}[m_2, \tilde\mu] + \\
& \frac{17}{12} g_2^4 \mathrm{LF}_{2,2,-1}[m_2, \tilde\mu] - \frac{1}{3} g_2^4 \mathrm{LF}_{3,1,-1}[m_2, \tilde\mu] - \frac{7}{3} g_2^4 \mathrm{LF}_{3,2,-2}[m_2, \tilde\mu] + \\
& \frac{3}{2} g_2^4 m_2^2 \mathrm{LF}_{3,2,-1}[m_2, \tilde\mu] + g_2^4 \mathrm{LF}_{3,3,-3}[m_2, \tilde\mu] - g_2^4 m_2^2 \mathrm{LF}_{3,3,-2}[m_2, \tilde\mu] + \\
& g_2^4 \mathrm{LF}_{4,2,-3}[m_2, \tilde\mu] - g_2^4 m_2^2 \mathrm{LF}_{4,2,-2}[m_2, \tilde\mu] - \frac{1}{4} g_2^2 \overline{a_e}^{pr} a_e^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde l}^p, m_{\tilde e}^r] + \\
& \frac{1}{6} g_2^2 \left(\overline{a_e}^{pr} a_e^{pr} + \tilde\mu^2 \overline{y_e}^{pr} y_e^{pr}\right) \mathrm{LF}_{5,1,-2}[m_{\tilde l}^p, m_{\tilde e}^r] - \frac{3}{4} g_2^2 \overline{a_d}^{pr} a_d^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde q}^p, m_{\tilde d}^r] + \\
& \frac{1}{2} g_2^2 \left(\overline{a_d}^{pr} a_d^{pr} + \tilde\mu^2 \overline{y_d}^{pr} y_d^{pr}\right) \mathrm{LF}_{5,1,-2}[m_{\tilde q}^p, m_{\tilde d}^r] + \frac{3}{4} \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{2,1,0}[m_{\tilde q}^p, m_{\tilde q}^s] - \\
& \frac{3}{4} \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{3,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s] - \frac{1}{4} g_2^2 \left(\overline{a_u}^{pr} a_u^{pr} + \tilde\mu^2 \overline{y_u}^{pr} y_u^{pr}\right) \mathrm{LF}_{2,2,0}[m_{\tilde q}^p, m_{\tilde u}^r] + \\
& \frac{3}{4} g_2^2 \left(-\overline{a_u}^{pr} a_u^{pr} + \tilde\mu^2 \overline{y_u}^{pr} y_u^{pr}\right) \mathrm{LF}_{3,1,0}[m_{\tilde q}^p, m_{\tilde u}^r] + \\
& \frac{1}{4} g_2^2 \left(2\,\overline{a_u}^{pr} a_u^{pr} - \tilde\mu^2 \overline{y_u}^{pr} y_u^{pr}\right) \mathrm{LF}_{3,2,-1}[m_{\tilde q}^p, m_{\tilde u}^r] + \\
& \frac{3}{4} g_2^2 \left(\overline{a_u}^{pr} a_u^{pr} - \tilde\mu^2 \overline{y_u}^{pr} y_u^{pr}\right) \mathrm{LF}_{4,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r] + \frac{3}{4} \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{2,1,0}[m_{\tilde q}^s, m_{\tilde q}^p] - \\
& \frac{3}{4} \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{3,1,-1}[m_{\tilde q}^s, m_{\tilde q}^p] + \frac{1}{4} g_2^2 \left(\overline{a_u}^{pr} a_u^{pr} + \tilde\mu^2 \overline{y_u}^{pr} y_u^{pr}\right) \mathrm{LF}_{3,1,0}[m_{\tilde u}^r, m_{\tilde q}^p] + \\
& \frac{1}{4} g_2^2 \left(\overline{a_u}^{pr} a_u^{pr} + \tilde\mu^2 \overline{y_u}^{pr} y_u^{pr}\right) \mathrm{LF}_{3,2,-1}[m_{\tilde u}^r, m_{\tilde q}^p] - \\
& \frac{3}{4} g_2^2 \left(\overline{a_u}^{pr} a_u^{pr} + \tilde\mu^2 \overline{y_u}^{pr} y_u^{pr}\right) \mathrm{LF}_{4,1,-1}[m_{\tilde u}^r, m_{\tilde q}^p] + \\
& \frac{1}{2} g_2^2 \left(\overline{a_u}^{pr} a_u^{pr} + \tilde\mu^2 \overline{y_u}^{pr} y_u^{pr}\right) \mathrm{LF}_{5,1,-2}[m_{\tilde u}^r, m_{\tilde q}^p] - \frac{3}{2} \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{2,1,0}[m_{\tilde u}^t, m_{\tilde d}^r] + \\
& \frac{3}{2} \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{3,1,-1}[m_{\tilde u}^t, m_{\tilde d}^r] + \frac{5}{12} g_1^2 g_2^2 \mathrm{LF}_{4,1,-2}[\tilde\mu, m_1] - \frac{1}{3} g_1^2 g_2^2 \mathrm{LF}_{5,1,-3}[\tilde\mu, m_1] - \\
& \frac{4}{3} g_2^4 \mathrm{LF}_{3,1,-1}[\tilde\mu, m_2] - \frac{5}{3} g_2^4 \mathrm{LF}_{3,2,-2}[\tilde\mu, m_2] + \frac{9}{4} g_2^4 \mathrm{LF}_{4,1,-2}[\tilde\mu, m_2] + \frac{1}{2} g_2^4 \mathrm{LF}_{4,2,-3}[\tilde\mu, m_2] - \\
& \frac{1}{2} g_2^4 \tilde\mu^2 \mathrm{LF}_{4,2,-2}[\tilde\mu, m_2] - g_2^4 \mathrm{LF}_{5,1,-3}[\tilde\mu, m_2] + 2 g_1^2 g_2^2 \mathrm{LF}_{2,2,1,-2}[m_2, \tilde\mu, m_1] + \\
& \frac{3}{2} m_1 m_2 g_1^2 g_2^2 \mathrm{LF}_{2,2,1,-1}[m_2, \tilde\mu, m_1] - g_1^2 g_2^2 \mathrm{LF}_{3,2,1,-3}[m_2, \tilde\mu, m_1] - \\
& m_1 m_2 g_1^2 g_2^2 \mathrm{LF}_{3,2,1,-2}[m_2, \tilde\mu, m_1] - \frac{1}{4} \tilde\mu^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde l}^p, m_{\tilde l}^s, m_{\tilde e}^r] + \\
& \frac{3}{4} \left(\overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} a_d^{sr} - \tilde\mu^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr}\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^r] + \\
& \frac{3}{4} \left(\overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} a_u^{sr} - \tilde\mu^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r] + \\
& \frac{3}{4} \left(-\overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} a_u^{sr} + \tilde\mu^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r] - \\
& \frac{3}{4} \tilde\mu^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^t] + \frac{3}{4} \tilde\mu^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \\
& \quad \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^t] + \frac{3}{4} \overline{y_d}^{st} y_u^{sr} a_d^{pt} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde d}^t] + \\
& \frac{3}{8} \left(-\overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} a_u^{sr} + \tilde\mu^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr}\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde q}^s] + \\
& \frac{3}{4} y_d^{sr} \overline{y_u}^{st} \left(\overline{a_d}^{pr} a_u^{pt} - \tilde\mu^2 \overline{y_d}^{pr} y_u^{pt}\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde u}^t, m_{\tilde d}^r] + \\
& \frac{3}{8} \tilde\mu^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde u}^t, m_{\tilde q}^s] + \frac{3}{4} \overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^r] - \\
& \frac{3}{4} \overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^r] - \frac{3}{8} \overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde q}^p] - \\
& \frac{3}{4} \tilde\mu^2 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^s, m_{\tilde u}^t, m_{\tilde d}^r] - \\
& \frac{3}{2} \overline{y_d}^{st} y_u^{sr} a_d^{pt} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde u}^r, m_{\tilde d}^t, m_{\tilde q}^p] + \frac{3}{2} \overline{y_d}^{st} y_u^{sr} a_d^{pt} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^r, m_{\tilde d}^t, m_{\tilde q}^p] + \\
& \frac{3}{2} y_d^{sr} \overline{y_u}^{st} \left(-\overline{a_d}^{pr} a_u^{pt} + \tilde\mu^2 \overline{y_d}^{pr} y_u^{pt}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^p] + \\
& \frac{3}{2} y_d^{sr} \overline{y_u}^{st} \left(\overline{a_d}^{pr} a_u^{pt} - \tilde\mu^2 \overline{y_d}^{pr} y_u^{pt}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^p] + \\
& \frac{3}{2} \tilde\mu^2 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{2,1,1,0}[m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^s] - \\
& \frac{3}{2} \tilde\mu^2 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^s] - \frac{3}{4} g_1^2 g_2^2 \mathrm{LF}_{2,1,1,-1}[\tilde\mu, m_1, m_2] + \\
& g_1^2 g_2^2 \mathrm{LF}_{3,1,1,-2}[\tilde\mu, m_1, m_2] - \frac{1}{2} g_1^2 g_2^2 \mathrm{LF}_{4,1,1,-3}[\tilde\mu, m_1, m_2] - g_1^2 g_2^2 \mathrm{LF}_{3,2,1,-3}[\tilde\mu, m_2, m_1] - \\
& m_1 m_2 g_1^2 g_2^2 \mathrm{LF}_{3,2,1,-2}[\tilde\mu, m_2, m_1] - \frac{1}{4} \tilde\mu^2 \overline{y_e}^{st} y_e^{pt} \overline{a_e}^{pr} a_e^{sr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde l}^p, m_{\tilde l}^s, m_{\tilde e}^r, m_{\tilde e}^t] - \\
& \frac{3}{4} \tilde\mu^2 \overline{y_d}^{st} y_d^{pt} \overline{a_d}^{pr} a_d^{sr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^r, m_{\tilde d}^t] + \\
& \frac{3}{4} \left(\overline{a_d}^{pr} a_d^{sr} \overline{a_u}^{st} a_u^{pt} + \tilde\mu^4 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^r, m_{\tilde u}^t] - \\
& \frac{3}{4} \tilde\mu^2 \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde u}^t] + \\
& \frac{3}{4} \tilde\mu^2 \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde u}^t] - \\
& \frac{3}{4} \tilde\mu^2 \overline{y_d}^{st} y_u^{sr} a_d^{pt} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde d}^t, m_{\tilde q}^s] + \\
& \frac{3}{8} \tilde\mu^2 \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde q}^s, m_{\tilde u}^t] + \\
& \frac{3}{4} \left(\tilde\mu^4 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} + \overline{a_d}^{pr} a_u^{pt} \left(a_d^{sr} \overline{a_u}^{st} - \tilde\mu^2 y_d^{sr} \overline{y_u}^{st}\right)\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^s] + \\
& \frac{3}{8} \tilde\mu^2 \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde u}^t, m_{\tilde q}^s, m_{\tilde u}^r] - \\
& \frac{3}{4} \tilde\mu^2 \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde u}^t] + \\
& \frac{3}{4} \tilde\mu^2 \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde u}^t] - \\
& \frac{3}{4} \tilde\mu^2 \overline{y_d}^{st} y_u^{sr} a_d^{pt} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde d}^t, m_{\tilde q}^p] + \\
& \frac{3}{8} \tilde\mu^2 \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde q}^p, m_{\tilde u}^t] + \\
& \frac{3}{4} \left(\tilde\mu^4 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} + \overline{a_d}^{pr} a_u^{pt} \left(a_d^{sr} \overline{a_u}^{st} - \tilde\mu^2 y_d^{sr} \overline{y_u}^{st}\right)\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^s, m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^p] + \\
& \frac{3}{8} \tilde\mu^2 \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^s, m_{\tilde u}^t, m_{\tilde q}^p, m_{\tilde u}^r] + \\
& \frac{3}{2} \tilde\mu^2 \overline{y_d}^{st} y_u^{sr} a_d^{pt} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde u}^r, m_{\tilde d}^t, m_{\tilde q}^p, m_{\tilde q}^s] - \\
& \frac{3}{2} \tilde\mu^2 \overline{y_d}^{st} y_u^{sr} a_d^{pt} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde u}^r, m_{\tilde d}^t, m_{\tilde q}^p, m_{\tilde q}^s] - \\
& \frac{3}{2} \left(\tilde\mu^4 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} + \overline{a_d}^{pr} a_u^{pt} \left(a_d^{sr} \overline{a_u}^{st} - \tilde\mu^2 y_d^{sr} \overline{y_u}^{st}\right)\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^p, m_{\tilde q}^s] + \\
& \frac{3}{2} \left(\tilde\mu^4 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} + \overline{a_d}^{pr} a_u^{pt} \left(a_d^{sr} \overline{a_u}^{st} - \tilde\mu^2 y_d^{sr} \overline{y_u}^{st}\right)\right) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^p, m_{\tilde q}^s] \Big)
\end{aligned}
$$